K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

Basics of Neural Network & Deep Learning

Theory Assignment-1

SUBMITTED BY:

Diwakar Gupta

2301201014

BCA (AI&DS)

SECTION -A

SUBMITTED TO:

Mr. Anoja Mishra

1. Explain the role of weights in a neuron.

Answer

1. Weights represent the importance of each input feature in a neuron. Each input is multiplied by its corresponding weight before being passed forward.
2. They control how much influence an input has on the final output. Higher weight means greater impact, lower weight means less impact.
3. Weights help the neuron learn patterns from data. During training, the model adjusts these values to improve accuracy.
4. They are updated using backpropagation and optimization techniques like gradient descent to reduce error.
5. Weights, along with bias, determine the final output of the neuron after applying the activation function.
   Properly learned weights allow the neural network to make accurate predictions and generalize well to new data.

2. What is an activation function?

Answer

An activation function is applied to the weighted sum of inputs (before producing the final output of a neuron). It introduces non-linearity, enabling the model to learn and represent complex data patterns. Without it, even a deep neural network would behave like a simple linear regression model. Activation functions decide whether a neuron should be activated based on the weighted sum of inputs and a bias term. They also make backpropagation possible by providing gradients for weight updates.

Types of Activation Functions

1. Linear Activation Function

Linear Activation Function resembles straight line define by y=x. No matter how many layers the neural network contains if they all use linear activation functions the output is a linear combination of the input.

- The range of the output spans from $(-\infty$ to $+\infty)$.
- Linear activation function is used at just one place i.e. output layer.
- Using linear activation across all layers makes the network's ability to learn complex patterns limited.

Linear activation functions are useful for specific tasks but must be combined with non-linear functions to enhance the neural network's learning and predictive capabilities.

Linear Activation Function or Identity Function returns the input as the output

2. Non-Linear Activation Functions

1. Sigmoid Function

It is characterized by 'S' shape. It is mathematically defined as$A = \frac{1}{1+e^{-x}}$. This formula ensures a smooth and continuous output that is essential for gradient-based optimization methods.

- It allows neural networks to handle and model complex patterns that linear equations cannot.
- The output ranges between 0 and 1, hence useful for binary classification.

2. Tanh Activation Function

It (hyperbolic tangent function) is a shifted version of the sigmoid, allowing it to stretch across the y-axis. It is defined as:

$$f(x) = \tanh(x) = \frac{2}{1 + e^{-2x}} - 1.$$

Alternatively, it can be expressed using the sigmoid function:

$$\tanh(x) = 2 \times \text{sigmoid}(2x) - 1$$

3. ReLU (Rectified Linear Unit) Function

It is defined by $A(x) = \max(0, x)$, this means that if the input x is positive, ReLU returns x, if the input is negative, it returns 0.

At a time only a few neurons are activated making the network sparse making it efficient and easy for computation.

ReLU Activation Function

4. Leaky ReLU

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

- It is similar to ReLU but allows a small negative slope (α, e.g., 0.01) instead of zero.
- Solves the "dying ReLU" problem, where neurons get stuck with zero outputs.
- Range: $(-\infty, \infty)$.
- Preferred in some cases for better gradient flow.

Leaky ReLU Activation Function

5. SoftPlus Function

It is defined mathematically as: $A(x) = \log(1 + e^x)$.

This equation ensures that the output is always positive and differentiable at all points which is an advantage over the traditional ReLU function.

3. Define probability distribution in ML context.

Answer

1. A probability distribution describes how the possible values of a variable are distributed along with their likelihood of occurrence.
2. In machine learning, it represents how data points are spread and how likely different outcomes are.
3. It helps models understand the underlying pattern of data and make predictions based on probabilities.
4. There are two main types: discrete probability distribution for countable values and continuous probability distribution for measurable values.
5. It is used in tasks like classification, regression, and uncertainty estimation.
6. Many machine learning algorithms assume a specific probability distribution of data to simplify learning and improve performance.

4. What is gradient in optimization?

Answer

is an optimisation algorithm used to reduce the error of a machine learning model. It works by repeatedly adjusting the model's parameters in the direction where the error decreases the most hence helping the model learn better and make more accurate predictions.

- Used to minimize the error or loss of a machine learning model
- Updates model parameters step by step using gradients
- Helps models like neural networks and regression learn better weights

Implementation

1. Creates a dataset: Generates 200 sample data points with some random noise.
2. Initializes parameters: Starts with initial values for weight and bias, and sets the learning rate and number of iterations.
3. Trains the model: In each iteration, the model makes predictions, calculates the error and updates the parameters using Gradient Descent.
4. Tracks the loss: Stores the loss value in every iteration to observe how the error changes.
5. Plots the loss graph: Shows how the error decreases over iterations.
6. Plots the fitted line: Displays the data points along with the final regression line learned by the model.